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SEAT No. _____

No. of Printed Pages : 2

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SARDAR PATEL UNIVERSITY
M. Sc. (IT) INTEGRATED EXAMINATION, VII SEM

Date : 20th OCT 2018, Saturday

Time : 2:00pm To 05:00pm

Session : Evening

Sub: System Software

Course No : PS07CIIT01

Total marks : 70

Q – 1 Select an appropriate option

[08]

- i) _____ part build the desired target program by using intermediate representation of two part.
 a) Analysis b) Synthesis c) Execution d) All of above
- ii) A program in execution is called .
 a) Process b) Instruction c) Procedure d) Function
- iii) An imperative statement indicates an action to be performed during the _____ of assembled program.
 a) Execution b) Initialization c) Implementation d) Declaration
- iv) Mnemonic has _____ primary fields.
 a) Name and Address b) Mnemonic and opcode
 c) Mnemonic and name d) Name and opcode
- v) The _____ maintains an internal form which is used to perform the edit operations.
 a) Editor b) Loader c) Linker d) None of these
- vi) _____ is the process of binding an external reference to the correct link time address.
 a) Translation b) Linking c) Loading d) None of these
- vii) _____ helps the user in selecting test data for his program.
 a) Program environment b) Language processor
 c) Software tools d) Test data generator
- viii) _____ helps in obtaining information for localization of errors.
 a) Translator b) Profile monitor c) Editors d) Debug Monitors

Q – 2 Answer the following Questions (Any SEVEN)

[14]

- i) Draw a diagram of a language processing system.
- ii) Define specification Gap and Execution Gap.
- iii) Differentiate between Assembler, Compiler and Interpreter.
- iv) List the tasks performed by the passes of a two pass assembler.
- v) What is Condition code? How it is tested?
- vi) Differentiate between linking and loading.
- vii) Why origin of a program may have to be changed by the linker or loader?
- viii) List down steps for testing and debugging.
- ix) Define Profile Monitor.

Q – 3 a) What is compiler? Explain its different phases.

[06]

b) Write short note on code optimization with its advantages.

[06]

OR

Q – 3 b) Explain software tools that manipulate source programs

[06]

Q – 4 (a) Describe Pass I assembler, Processing of Declaration and Assembler directives.

[06]

(b) What is assembly language? Explain the three basic facilities of assembly language.

[06]

OR

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Q – 4 (b) What is an expression tree? How an expression is evaluated using an expression tree? Discuss its advantages over the other evaluation techniques. [06]

Q – 5 a) Discuss various steps for execution of program written in a programming language. [06]

b) Explain Translated, Linked and Load time address. [06]

OR

Q – 5 b) Explain different editors in detail. [06]

Q – 6 a) Explain debug monitor with how to produce debug information [06]

b) Which are the four basic categories in which programming environment can be classified? Explain each in brief. [06]

OR

Q – 6 b) What is software tools? Discuss different software tools for programming development. [06]

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(2)